

**SOFTWARE REQUIREMENTS SPECIFICATION
FOR
SKILLBRIDGE AI**

PREPARED BY

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Table of Contents

Table of Contents	0
Revision History	0
1.1 Introduction	0
1.2 Purpose	0
1.3 Document Conventions	0
1.4 Intended Audience and Reading Suggestions	0
1.5 Product Scope	0
1.6 References	0
2. Overall Description	0
2.1 Product Perspective	0
2.2 Product Functions	0
2.3 User Classes and Characteristics	0
2.4 Operating Environment	0
2.5 Design and Implementation Constraints	0
2.6 User Documentation	0
2.7 Assumptions and Dependencies	0
3. External Interface Requirements	0
3.1 User Interfaces	0
3.2 Hardware Interfaces	0
3.3 Software Interfaces	0
3.4 Communications Interfaces	0
4. System Features	0
4.1 System Feature 1	0
4.2 System Feature 2 (and so on)	0
5. Other Nonfunctional Requirements	0
5.1 Performance Requirements	0
5.2 Safety Requirements	0
5.3 Security Requirements	0
5.4 Software Quality Attributes	0
5.5 Business Rules	0
6. Other Requirements	0
Appendix A: Glossary	0
Appendix B: Analysis Models	0
Appendix C: To Be Determined List	0

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

SkillBridge AI is an enterprise web application intended to help organizations identify employees with suitable skills for internal projects. In many companies, employee expertise is distributed across departments and stored in resumes, emails, or fragmented HR records, making it difficult for managers to discover the right people when new project requirements arise. The proposed system addresses this problem by allowing employees to upload resumes and maintain professional profiles while AI-based services automatically extract relevant skill information from those documents.

The purpose of this Software Requirements Specification is to define the behavior, scope, interfaces, constraints, and quality expectations of SkillBridge AI in a clear and verifiable form. The document provides a common reference for project stakeholders such as developers, testers, reviewers, project guides, and administrators. It is intended to support implementation planning, requirement validation, testing, and final-year academic project documentation.

1.2 Document Conventions

This document follows an IEEE-style SRS structure and uses hierarchical section numbering. Functional requirements are uniquely identified using labels such as REQ-4.1.1 and are written using the term “shall” to indicate mandatory system behavior. Chapters are organized into Introduction, Overall Description, External Interface Requirements, System Features, Non-Functional Requirements, and Other Requirements. Technical terms such as employee, manager, HR user, administrator, parsing engine, and recommendation engine are used consistently throughout the document.

1.2 Intended Audience

This SRS is intended for multiple stakeholder groups. Developers and software engineers use it as a baseline for implementing the frontend, backend, AI modules, and database interactions. Testers derive validation scenarios and acceptance tests from the listed requirements. Project managers and academic reviewers use it to understand the scope, structure, and expected outcomes of the project. Administrators and future maintainers can use it as a reference for operational controls, security expectations, and system maintenance responsibilities.

1.3 Product Scope

SkillBridge AI provides a centralized skill discovery platform that helps organizations match internal talent with project requirements. Employees upload resumes in supported formats such as PDF and DOCX, and the system applies Artificial Intelligence, Natural Language Processing, and OCR-based techniques to extract skills, certifications, experience, education, and project details. Managers can then search employees or enter project requirements and obtain ranked recommendations of the most suitable candidates. The system reduces manual resume screening, improves workforce visibility, and supports faster, more data-driven staffing decisions.

1.4 References

- IEEE-style Software Requirements Specification practices and terminology.
- Technology references for React.js, Node.js, Express.js, MongoDB, and spaCy.
- Agile development practices used for incremental implementation and review.

2. Overall Description

2.1 Product Perspective

SkillBridge AI is designed as a browser-based enterprise platform that sits between employee profile management and project staffing activities. Instead of depending on manual resume review or informal knowledge of available talent, the system maintains a searchable organizational skill repository generated from uploaded resumes and profile data. The application can function as a standalone internal solution or be extended later to integrate with HRMS, identity systems, or project management platforms through APIs.

From an architectural perspective, the system consists of a responsive web frontend, backend service layer, MongoDB-based persistence layer, resume parsing pipeline, and recommendation module. Employees contribute resumes and profile updates, the parsing engine converts unstructured documents into structured records, and managers or HR users consume that information through search, filtering, and recommendation workflows. This structure supports modular development and allows the project to be implemented iteratively under Agile methodology.

2.2 Product Functions

The main functions of SkillBridge AI include secure user authentication, employee profile management, resume upload and validation, AI-based resume parsing, skill extraction, project requirement management, smart employee search, AI-based employee recommendation, dashboards, notifications, and administrative management. Employees can maintain their profiles and resumes, managers can define staffing requirements and review recommended candidates, HR users can monitor workforce information, and administrators can manage users, roles, logs, and system-level settings.

2.3 User Classes and Characteristics

The system includes four primary user classes: Employees, Project Managers, HR Personnel, and Administrators. Employees are responsible for uploading resumes, updating profile details, and reviewing extracted skill information. Project Managers create project requirements, search for employees with specific competencies, and use AI-generated recommendations for staffing decisions. HR Personnel oversee employee information quality, support profile maintenance, and generate workforce reports. Administrators handle privileged operations such as user management, role assignment, audit log review, backup controls, and system configuration.

2.4 Operating Environment

SkillBridge AI operates as a responsive web application accessible from modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The frontend may be implemented using React.js with Tailwind CSS for a dashboard-style interface. Backend services may be implemented using Node.js and Express.js, while MongoDB is used for storing user records, employee profiles, parsed resume data, project requirements, notifications, and logs. Resume parsing and OCR functionality may be implemented using Python libraries such as spaCy, PyMuPDF, and OCR tools for scanned documents.

2.5 Design Constraints

The application shall enforce secure authentication and role-based authorization for all protected operations. Resume uploads shall be restricted to approved file types and validated before processing. Because the project is intended for a final-year academic submission while still reflecting industry practice, the design should remain modular, readable, and maintainable rather than excessively complex. The system should also support responsive UI design, secure

communication over HTTPS in production, and separation of concerns between UI, business logic, AI services, and persistence.

2.6 User Documentation

The project should include role-oriented user documentation such as an Employee User Guide, a Manager Guide for search and recommendation workflows, and an Administrator Guide for user management and operational tasks. In addition, the application should provide contextual help through labels, validation messages, tooltips, and clear status indicators so that first-time users can understand the workflow without extensive training.

2.7 Assumptions and Dependencies

The system assumes that employees upload accurate and sufficiently detailed resumes and that managers define project requirements with meaningful skill and experience criteria. Recommendation quality depends on the completeness of extracted profile data and the quality of project requirements. The system also depends on MongoDB or an equivalent database, resume parsing libraries, OCR support for scanned resumes, secure internet connectivity, and notification services if email alerts are enabled.

3. External Interface Requirements

3.1 User Interface

SkillBridge AI shall provide a modern, responsive, and role-aware web interface designed for enterprise use. Employees shall have access to screens for registration, login, dashboard overview, resume upload, profile editing, and extracted skill review. Managers shall have access to project requirement forms, employee search pages, recommendation views, and project tracking screens. HR users and administrators shall have access to management dashboards, analytics, profile review pages, and administrative controls based on authorization.

The interface should follow contemporary UI/UX principles by providing sidebar navigation, dashboard cards, validation messages, confirmation dialogs, notifications, loading indicators, and search filters. The design should minimize the number of clicks needed for core tasks and provide clear feedback during long-running operations such as resume parsing or

recommendation generation. Where appropriate, the interface may support dark mode, progress indicators, and accessible color contrast for better usability.

3.2 Hardware Interface

The application does not require specialized end-user hardware beyond a desktop, laptop, tablet, or smartphone with internet connectivity and a modern browser. On the server side, the system requires sufficient CPU, memory, and storage capacity to support concurrent user requests, file uploads, parsing operations, recommendation generation, and database queries. If OCR processing is enabled for scanned documents, the infrastructure should provide adequate resources to handle those additional workloads without significantly affecting response time.

3.3 Software Interface

The frontend communicates with backend services through RESTful APIs using JSON payloads. The backend interfaces with MongoDB for persistence of employee records, project requirements, extracted skills, notifications, and logs. AI-related software interfaces include a resume parsing module that extracts structured information from uploaded resumes and an optional OCR layer for image-based or scanned documents. Notification interfaces may connect to email services for sending status updates, while cloud storage services may be used to store uploaded resumes securely.

3.4 Communication Interface

Communication between the client and server shall occur over HTTP/HTTPS, with HTTPS required for production deployment. File uploads shall use multipart/form-data, while other operations such as authentication, search, project requirement management, and recommendation retrieval shall use JSON-based request and response bodies. Authentication tokens or secure sessions shall be validated on each protected request. Error responses shall be structured clearly so that the frontend can display actionable messages to the user when validation fails, a resource is missing, or a dependent service becomes unavailable.

4. System Features

4.1 User Authentication and Authorization

Description and Priority: Provides secure registration, login, logout, password management, and role-based access to all protected parts of the application. Priority: High.

Functional Requirements:

- REQ-4.1.1 The system shall allow users to register with mandatory profile information such as name, email, employee identifier, department, and password.
- REQ-4.1.2 The system shall authenticate users before granting access to protected resources.
- REQ-4.1.3 The system shall store passwords using a secure hashing mechanism such as BCrypt.
- REQ-4.1.4 The system shall enforce role-based access for Employees, Managers, HR users, and Administrators.
- REQ-4.1.5 The system shall provide logout functionality and a password reset workflow.

4.2 Employee Profile Management

Description and Priority: Allows employees to create and maintain professional profile information that complements parsed resume data. Priority: High.

Functional Requirements:

- REQ-4.2.1 The system shall allow employees to view and edit profile fields such as designation, department, experience summary, skills, certifications, and contact details as permitted by policy.
- REQ-4.2.2 The system shall maintain a profile record for each employee account.
- REQ-4.2.3 The system shall allow HR or authorized administrators to review or update employee profile information when required.
- REQ-4.2.4 The system shall display the latest profile information and update timestamps to authorized users.

4.3 Resume Upload and Validation

Description and Priority: Allows employees to upload resumes that are validated before storage and AI processing. Priority: High.

Functional Requirements:

- REQ-4.3.1 The system shall accept resumes in supported formats such as PDF and DOCX.
- REQ-4.3.2 The system shall validate file type, file size, and basic file integrity before accepting an upload.
- REQ-4.3.3 The system shall associate each uploaded resume with the correct employee profile.
- REQ-4.3.4 The system shall store upload and processing status for each resume.

4.4 Resume Parsing and Skill Extraction

Description and Priority: Extracts structured information from resumes and stores it as searchable employee data. Priority: High.

Functional Requirements:

- REQ-4.4.1 The system shall extract skills, education, certifications, projects, and work experience from uploaded resumes.
- REQ-4.4.2 The system shall store extracted data in structured employee profile fields or linked parsing records.
- REQ-4.4.3 The system shall mark parsing status as pending, completed, failed, or review required.
- REQ-4.4.4 The system shall allow manual review or correction of extracted data where confidence is low or extraction is incomplete.

4.5 Smart Employee Search

Description and Priority: Allows managers and HR users to search employees using skill-based and profile-based criteria. Priority: High.

Functional Requirements:

- REQ-4.5.1 The system shall support search by skill, technology, certification, experience, department, and employee name.
- REQ-4.5.2 The system shall support multi-criteria filtering in a single search operation.
- REQ-4.5.3 The system shall display matching employees sorted or ranked by relevance to the query.
- REQ-4.5.4 The system shall allow authorized users to open detailed employee profiles from search results.

4.6 Project Requirement Management

Description and Priority: Allows managers to create and maintain project staffing requirements for recommendation and tracking purposes. Priority: High.

Functional Requirements:

- REQ-4.6.1 The system shall allow authorized users to create, edit, view, and delete project requirement records.
- REQ-4.6.2 Each requirement shall support fields such as project name, required skills, experience level, certifications, and project status.
- REQ-4.6.3 The system shall validate mandatory fields before saving a requirement.

4.7 AI-Based Employee Recommendation

Description and Priority: Compares project requirements with employee profiles and recommends suitable employees. Priority: High.

Functional Requirements:

- REQ-4.7.1 The system shall calculate employee suitability based on skill match, experience, certifications, and other relevant criteria.
- REQ-4.7.2 The system shall generate a ranked list of recommended employees for a selected project requirement.
- REQ-4.7.3 The system shall allow managers to open detailed employee profiles from recommendation results.
- REQ-4.7.4 The system shall support re-running recommendations when project requirements change.

4.8 Dashboard and Analytics

Description and Priority: Provides visual summaries of workforce skill data, profile completeness, and project staffing activity. Priority: Medium.

Functional Requirements:

- REQ-4.8.1 The system shall display dashboard widgets relevant to the logged-in user role.
- REQ-4.8.2 The system shall provide analytics such as skill distribution, active requirements, pending reviews, or recommendation counts.
- REQ-4.8.3 The system shall restrict dashboard visibility based on authorization rules.

4.9 Notifications

Description and Priority: Notifies users about important workflow events such as parsing results and staffing-related actions. Priority: Medium.

Functional Requirements:

- REQ-4.9.1 The system shall generate notifications for events such as resume upload completion, parsing failure, profile review request, and recommendation availability.
- REQ-4.9.2 The system shall support in-app notifications and may support email notifications where configured.
- REQ-4.9.3 The system shall allow users to view notification history relevant to their account.

4.10 Administration Module

Description and Priority: Supports privileged administrative tasks such as user management, role management, audit review, and operational controls. Priority: High.

Functional Requirements:

- REQ-4.10.1 The system shall allow administrators to manage user accounts and assign roles.
- REQ-4.10.2 The system shall maintain audit logs for critical actions and security-related events.
- REQ-4.10.3 The system shall allow administrators to manage selected system settings and maintenance controls, including backup-related operations where implemented.

5. Non-Functional Requirements

5.1 Performance

The system should provide acceptable response times for common operations such as login, profile retrieval, employee search, and project requirement management under expected academic project load conditions. Resume parsing and recommendation generation may take longer than standard CRUD operations, but the application should provide progress indicators or status updates for long-running tasks. Search operations should be optimized through appropriate indexing and efficient query design so that managers can retrieve results without unnecessary delay.

5.2 Security

SkillBridge AI shall protect user credentials and employment-related data through authentication, authorization, password hashing, and secure communication. Access to employee profiles, recommendation results, and administrative features shall be restricted according to user roles. Audit logs should capture important security-related events such as failed logins, role changes, and administrative actions. Sensitive data should not be exposed in error messages or client-side logs.

5.3 Reliability and Availability

The application should preserve data integrity during resume uploads, profile updates, and project requirement modifications. If a parsing or recommendation process fails, the system should record the failure and allow retry or follow-up review instead of silently losing the request. Within the scope of a final-year project, the system should be designed to remain consistently available during normal use, subject to hosting and infrastructure availability.

5.4 Scalability and Maintainability

The design should support modular growth so that new recommendation logic, additional profile fields, or integration endpoints can be introduced without major restructuring. Maintainability should be supported through separation of concerns between the UI, backend services, database layer, and AI processing modules. Clear naming, reusable APIs, and consistent validation rules should reduce the effort required for debugging and future enhancement.

5.5 Usability, Accessibility, and Portability

The interface should remain usable for employees, managers, HR users, and administrators with minimal training. Forms should provide clear labels, validation messages, and error handling so that users can recover from mistakes easily. The application should be accessible from standard modern browsers across common operating systems and device types. Color contrast, readable typography, and keyboard-friendly controls should be considered to improve accessibility and user experience.

6. Other Requirements

6.1 Database and Logging Requirements

The database shall store user accounts, employee profiles, uploaded resume references, extracted skill records, project requirements, recommendation results, notifications, and audit logs in a structured and queryable form. Logging should capture significant application events such as authentication attempts, resume parsing outcomes, administrative actions, and system errors. These records support troubleshooting, operational monitoring, and accountability.

6.2 Backup, API, and Future Scope

Where backup functionality is included, the system should support periodic protection of core data such as user accounts, employee profiles, and project requirements. Backend APIs should follow a consistent REST-oriented design so that frontend components and future integrations can consume the same services reliably. Future enhancements may include deeper HRMS integration, advanced recommendation scoring, skill gap analytics, team composition suggestions, and real-time notifications or collaboration features.